

Consistency-Constrained Multi-Task Learning for Bengali Hate Speech Classification & Generative Explanation

6-Week Research Execution Plan, Smoke Testing Pipeline & TigerLLM Benchmarking

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Senior Advisor Guidance Integrated

Target Venue: ICCIT 2026 (Cox's Bazar) / IEEE Access

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Executive Summary & Core Motivation

The Core Challenge

- Online hate speech in Bengali is **multi-dimensional** (*Hate Type, Target, Severity*).
- Existing Multi-Task Models suffer from **logical inconsistency**:
 - Predicting Type = None, but Target = Community.
 - Predicting Type = None, but Severity = Severe.
- Existing explanation datasets (e.g., BanHADEX) **lack severity conditioning** and use incompatible taxonomies.

Our Proposed Solution

- **Backbone**: Fine-tuned csebuetnlp/banglabert encoder.
- **Multi-Task Heads**: 3 classification heads + 1 generative decoder head.
- **Novel Loss ($\mathcal{L}_{\text{consist}}$)**: Differentiable soft penalty penalizing contradictory prediction vectors.
- **Generative Explanations**: Conditioned on all 3 target labels using LLM Silver-Standard annotations.

Taxonomy Mapping & Dataset Strategy

Dimension	BanglaMultiHate (Primary 50K)	BanHADEX (Reference 19K)
Hate Type	Abusive, Sexism, Religious Hate, Political Hate, Profane, None	Political, Religious, Gender, Personal, Abusive, Origin, Body Shaming
Target	Individual, Organization, Community, Society, None	7 specific target groups
Severity	Little to None, Mild, Severe	Not Present (Major Gap)
Explanations	Generated via Silver Pipeline (Gemini/TigerLLM)	Human-written (Unconditioned on severity)

Table: Taxonomy schema comparison justifying Silver-Standard Explanation Generation.

Key Research Takeaway

BanHADEX explanations cannot be directly reused because they lack severity conditioning. Therefore, we construct 5,000 Silver-Standard Bengali explanations conditioned on the full 3-task taxonomy.

Mathematical Formulation: Soft Consistency Loss

Logical Rules Enforced

- 1 If Type = None $\implies$ Target = None AND Severity = Little to None.
- 2 Contrapositive: If Target $\neq$ None OR Severity $\neq$ Little to None $\implies$ Type $\neq$ None.

Differentiable Consistency Loss ($\mathcal{L}_{\text{consist}}$)

Let $P_{\text{type}}(\text{None})$, $P_{\text{target}}(\neq \text{None})$, $P_{\text{sev}}(\neq \text{Little})$, and $P_{\text{sev}}(\text{Severe})$ be soft output probabilities:

$$\mathcal{L}_{\text{consist}} = \frac{1}{N} \sum_{i=1}^N [P_i(\text{type}=\text{None}) \cdot P_i(\text{target} \neq \text{None}) + P_i(\text{type}=\text{None}) \cdot P_i(\text{sev} \neq \text{Little}) + 2 \cdot P_i(\text{type}=\text{None}) \cdot P_i(\text{sev}=\text{Severe})]$$

- Fully differentiable formulation allows direct gradient backpropagation during fine-tuning.
- Dynamic weighting $\lambda(e)$ warms up linearly over epochs e .

Senior's Guidance 1: Mandatory Pipeline Smoke Test

Why Senior Recommended Smoke Testing First

Running a ****5-minute Smoke Test on 20 dummy samples**** before generating 5,000 API explanations or launching 15-epoch training loops prevents wasted GPU time, API costs, or Colab crashes from tensor shape mismatches and loss NaNs.

```
# === smoke_test.py Snippet ===
import torch
from src.dataset import BanglaHateDataset
from src.model import ConsistencyConstrainedMTL
from src.losses import ConsistencyPenaltyLoss

# 1. Load dummy dataset (10 samples)
loader = torch.utils.data.DataLoader(BanglaHateDataset('data/dummy_smoke_test.csv'), batch_size=2)
batch = next(iter(loader))

# 2. Test forward pass & backpropagation
model = ConsistencyConstrainedMTL('csebuethlp/banglabert')
optimizer = torch.optim.AdamW(model.parameters(), lr=1e-4)

type_out, target_out, sev_out, _ = model(batch['input_ids'], batch['attention_mask'])
loss = ConsistencyPenaltyLoss()(type_out, target_out, sev_out)
loss.backward()
optimizer.step()
```

Senior's Guidance 2: TigerLLM Integration (ACL 2025)

What is TigerLLM?

- Open-source Bengali Large Language Model (**ACL 2025**, Raihan et al.).
- Available in ****1B and 9B**** parameter sizes (csebuetnlp/TigerLLM-9B).
- Instruction-tuned on **Bangla-Instruct** (100K high-quality pairs).

Role 1: Local Silver Generator

- Replaces/complements Gemini 1.5 Pro.
- Runs locally on Kaggle/Colab GPU for ****free**** with zero rate limits.

Role 2: SOTA Open-Source LLM Baseline

- Evaluated in zero-shot / few-shot mode on hate speech test set.
- Answers reviewer critique: *"Why fine-tune BanglaBERT instead of prompting a Bengali LLM?"*

Role 3: Automated Explanation Judge

- Serves as an LLM-as-a-judge to evaluate explanation quality and fluency.

Experiment Matrix (4 BanglaBERT Ablations + 3 LLM Baselines)

Exp ID	Model Backbone	Consistency $\mathcal{L}_{\text{consist}}$	Gen Decoder $\mathcal{L}_{\text{gen}}$	Research Purpose
exp0_tiger	TigerLLM-9B	—	—	SOTA Bengali LLM Baseline (Zero-Shot / Few-Shot)
exp0_llama	Llama-3.2-3B	—	—	Multilingual LLM Baseline (BLP-2025 style)
exp1_base	BanglaBERT	No	No	Standard MTL Baseline
exp2_cons	BanglaBERT	Yes	No	Isolate Consistency Penalty Contribution
exp3_gen	BanglaBERT	No	Yes	Isolate Explanation Generation Contribution
exp4_full	BanglaBERT	Yes	Yes	Our Full Proposed Architecture

Table: Complete experimental setup to prove both classification and consistency gains.

Comprehensive Evaluation Strategy

1. Classification

- Macro F_1 per task:
 - F_1^{Type}
 - F_1^{Target}
 - F_1^{Severity}
- Average Macro F_1 .

2. Consistency Violation

- ****Consistency Violation Rate (CVR)****:

$$\text{CVR} = \frac{\text{Violating Predictions}}{\text{Total Predictions}}$$

- Key Goal: Lower CVR without sacrificing Macro F_1 .

3. ERASER Faithfulness

- ****Comprehensiveness (Comp $\uparrow$)****: Sufficiency drop when rationale removed.
- ****Sufficiency (Suff $\downarrow$)****: Prediction strength using rationale only.
- ****AOPC Score****.

6-Week Execution Roadmap

Week	Timeline	Key Focus & Deliverables
Week 1	Aug 19 – Aug 25	Environment Setup, EDA, Taxonomy Mapping & Pipeline Smoke Test
Week 2	Aug 26 – Sep 01	Silver Explanation Generation (Gemini 1.5 Pro + TigerLLM) & Custom Data Scraper
Week 3	Sep 02 – Sep 08	Model Architecture (model.py, losses.py) & Custom Dataset Annotation
Week 4	Sep 09 – Sep 15	Training 4 BanglaBERT Ablations + TigerLLM / Llama Baselines
Week 5	Sep 16 – Sep 22	Evaluation (Classification F_1 , CVR, ERASER Faithfulness Metrics)
Week 6	Sep 23 – Sep 30	Paper Writing, High-Res Figures, Proofreading & Final Submission

Team Task Division (Angkon & Nahid)

Phase	Angkon	Nahid
Week 1	Dataset Download & Comprehensive EDA	Environment Setup, BanglaBERT & Smoke Test
Week 2	Silver Explanation Prompting (Gemini API)	TigerLLM Local Setup + Custom Scraping
Week 3	Architecture Coding (model.py, losses.py)	Dataset Loader (dataset.py) + Annotation Lead
Week 4	Run Ablations 1 & 2 (exp1, exp2)	Run Ablations 3 & 4 + LLM Baselines
Week 5	Classification Metrics + CVR Analysis	ERASER Faithfulness Evaluation
Week 6	Write §4-5 (Setup, Results & Discussion)	Write §6-7 (Limitations, Conclusion, Figures)

Venue Strategy & Risk Management

Venue Options

- **Primary**: **ICCIT 2026** (Cox's Bazar, Dec 18-20). Deadline: Aug 31 / Extended Sep.
- **Backup 1**: **IEEE Access** (Journal — Rolling deadline, high impact).
- **Backup 2**: **BLP Workshop @ EMNLP 2027** (Top NLP venue).

Risk Mitigation Plan

- **Compute OOM**: Use batch size 8 + gradient accumulation; optional `banglabert_small`.
- **API Rate Limits**: Checkpointing every 100 samples; fallback to local 'TigerLLM-9B'.
- **Trade-off Risk**: If $\mathcal{L}_{\text{consist}}$ drops F_1 slightly, CVR reduction remains a strong paper contribution.

Summary & Immediate Next Steps

Current Status

- Research design and mathematical loss formulation finalized.
- Senior's feedback integrated (Smoke Testing + TigerLLM).
- Workspace plan synchronized in `implementation_plan2.md`.

Action Items for Today (Day 1)

- 1 Execute `smoke_test.py` on 10 dummy samples to verify PyTorch pipeline.
- 2 Test TigerLLM-9B vs Gemini 1.5 Pro on 20 samples for Bengali explanation quality.
- 3 Set up GitHub repository structure (`banglaHate/src`, `data`, `models`).

Thank You! Questions & Feedback?